

ANSWER KEY

MIDTERM
FALL 2024

STAT 642 STATISTICAL INFERENCE
INSTRUCTOR: GARRITT PAGE

Show all work to receive credit. Credit (including partial credit) is given for work not final answers.

1. A bag contains one marble which is either green or blue, with equal probabilities. A green marble is put into the bag (so there are two marbles now), and then a random marble is taken out. The marble taken out is green. What is the probability that the remaining marble is also green?

Let A = first marble pulled is green,

B = second marble pulled is green,

X = bag's first marble is green.

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$= \frac{P(A \cap B | X) + P(A \cap B | X^c)}{P(A | X) + P(A | X^c)}$$

$$= \frac{P(A \cap B \cap X)P(X) + P(A \cap B \cap X^c)P(X^c)}{P(A \cap X)P(X) + P(A \cap X^c)P(X^c)}$$

$$= \frac{1 \left(\frac{1}{2}\right) + 0 \left(\frac{1}{2}\right)}{1 \left(\frac{1}{2}\right) + \frac{1}{2} \left(\frac{1}{2}\right)} = \frac{\frac{1}{2}}{\frac{3}{4}} = \boxed{\frac{2}{3}}$$

2. Let $X \sim N(0, 1)$. Find the pdf of $Y = X^4$.

$$Y = X^4 ; \quad X = Y^{1/4} ; \quad dx = \frac{1}{4} Y^{-3/4} dy$$

$$\text{Support: } -\infty < Y^{1/4} < \infty ; \quad 0 < Y < \infty$$

$$\text{Let } A_1: -\infty < X < 0 \text{ and } A_2: 0 < X < \infty$$

$$\text{On } A_1, X = Y^{1/4} \quad \text{on } A_2, X = -Y^{1/4}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} (Y^{1/4})^2} \left| \frac{1}{4} Y^{-3/4} \right| + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} (-Y^{1/4})^2} \left| \frac{1}{4} Y^{-3/4} \right|$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} Y^{1/2}} \left| \frac{1}{4} Y^{-3/4} \right| + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} Y^{1/2}} \left| \frac{1}{4} Y^{-3/4} \right|$$

$$= \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2} Y^{1/2}} \cdot Y^{-3/4}, \quad 0 < Y < \infty$$

3. Let $Y = X + 1$ where $X \sim \text{Poisson}(\lambda)$. Find $E(Y^2)$.

$$\begin{aligned} E(Y^2) &= E((X+1)^2) = E(X^2 + 2X + 1) \\ &= E(X^2) + 2E(X) + 1 \\ &= [V(X) + E(X)^2] + 2E(X) + 1 \end{aligned}$$

$$\star E(X) = V(X) = \lambda$$

$$= \lambda + \lambda^2 + 2\lambda + 1$$

$$\boxed{= \lambda^2 + 3\lambda + 1}$$

4. Is it possible to have a random variable X such that the $E(X) = 0$, $E(X^2) = 1$, $E(X^3) = 0$, $E(X^4) = 0$?
Prove your claim.

No.

$$V(X^2) = E(X^4) - [E(X^2)]^2$$

In this example,

$$E(X^4) = 0 \text{ and } [E(X^2)]^2 = 1$$

$$\Rightarrow$$

$$V(X) = -1$$

This cannot be.

④

~~3. Let $Y = X + 1$ where $X \sim \text{Poisson}(\lambda)$. Find $E(Y^2)$.~~

OR By Jensen's, $E(g(X)) > g(E(X))$ if $g(\cdot)$ is convex.

Let X^2 be our r.v. and $g(x) = x^2$,
Note that $g(x)$ is convex.

$$E((X^2)^2) > [E(X^2)]^2$$

$$\Rightarrow E(X^4) > [E(X^2)]^2$$

$$\Rightarrow 0 > 1.$$

Contradiction!

~~4. Is it possible to have a random variable X such that the $E(X) = 0$, $E(X^2) = 1$, $E(X^3) = 0$, $E(X^4) = 0$?
Prove your claim.~~

5. Consider a random variable, X , with pdf

$$f(x) = \begin{cases} 4x & 0 < x < \frac{1}{2} \\ 4(1-x) & \frac{1}{2} \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_0^x 4x \, dx = 2x^2 \Big|_0^x = 2x^2$$

$$\frac{1}{2} + \int_{1/2}^x 4 - 4x \, dx = \frac{1}{2} + [4x - 2x^2]_{1/2}^x$$

Find $F_X(x)$, the cdf of X , and prove that it is a valid CDF. $= \frac{1}{2} + 4x - 2x^2 - 2 + \frac{1}{2} = -2x^2 + 4x - 1$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 2x^2 & 0 < x < 1/2 \\ -2x^2 + 4x - 1 & 1/2 \leq x < 1 \\ 1 & \text{o.w.} \end{cases}$$

i) $\lim_{x \rightarrow -\infty} F_X(x) = 0$ by construction

$\lim_{x \rightarrow \infty} F_X(x) = 1$ by construction

$$ii) \frac{d}{dx} F_X(x) = f_X(x) = \begin{cases} 4x & 0 < x < 1/2 \\ 4(1-x) & 1/2 \leq x < 1 \\ 0 & \text{o.w.} \end{cases}$$

$4x > 0$ on $x \in (0, 1/2)$, $4(1-x) > 0$ on $x \in [1/2, 1)$.

Thus, $F_X(x)$ is increasing.

iii) $2x^2$ and $-2x^2 + 4x - 1$ are polynomials and thus continuous. Check endpoints.

$$F_X(1/2) = 2(1/2)^2 = \frac{1}{2} = -2(1/2)^2 + 4(1/2) - 1 = -1/2 + 2 - 1 = 1/2 \checkmark$$

6. Let $\mathcal{F}$ be a σ -algebra on the sample space S . Prove that $\mathcal{F}$ is closed under countable intersection.

$\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ by definition (closed under union)

$\bigcup_{i=1}^{\infty} A_i^c \in \mathcal{F}$ by definition (closed under complement)

$\bigcup_{i=1}^{\infty} A_i^c = \left(\bigcap_{i=1}^{\infty} A_i \right)^c \in \mathcal{F}$ by De Morgan's Law

$\bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$ by definition (closed under complement)

7. Let X have a continuous cdf $F_X(x)$ and define the random variable Y as $Y = F_X(X)$. Show that $Y \sim UN(0,1)$. That is, show that $F_Y(y) = y$.

$$X = F_X^{-1}(Y)$$

$$F_Y(y) = P(Y \leq y)$$

$$= P(F_X(X) \leq y)$$

$$= P(X \leq F_X^{-1}(y))$$

$$= F_X(F_X^{-1}(y)) = y.$$

8. Consider a random variable X with PDF,

$$f_X(x) = \begin{cases} \frac{x}{4} & 0 < x \leq 2, \\ e^{-2(x-2)} & x > 2 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X)$ and $Var(X)$.

D	I
x	$e^{-2(x-2)}$
1	$-\frac{1}{2}e^{-2(x-2)}$
0	$\frac{1}{4}e^{-2(x-2)}$

$$E(X) = \int_0^2 x \cdot \frac{x}{4} dx + \int_2^{\infty} x e^{-2(x-2)} dx$$

$$= \left[\frac{1}{4} \cdot \frac{x^3}{3} \right]_0^2 + \left[-\frac{1}{2} x e^{-2(x-2)} - \frac{1}{4} e^{-2(x-2)} \right]_2^{\infty}$$

$$= \frac{8}{12} + \left[0 - 0 - \left(-1 - \frac{1}{4} \right) \right] = \frac{2}{3} + \frac{5}{4} = \boxed{\frac{23}{12}}$$

$$V(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_0^2 x^2 \cdot \frac{x}{4} dx + \int_2^{\infty} x^2 e^{-2(x-2)} dx$$

$$= \left[\frac{1}{4} \cdot \frac{x^4}{4} \right]_0^2 + \left[-\frac{1}{2} x^2 e^{-2(x-2)} - \frac{2}{4} x e^{-2(x-2)} - \frac{2}{18} e^{-2(x-2)} \right]_2^{\infty}$$

$$= 1 + \left[0 - 0 - 0 - \left(-2 - 1 - \frac{1}{4} \right) \right] = 1 + \frac{13}{4} = \frac{17}{4}$$

$$\text{So, } V(X) = \frac{17}{4} - \left[\frac{23}{12} \right]^2 = \boxed{\frac{83}{144}}$$

D	I
x^2	$e^{-2(x-2)}$
$2x$	$-\frac{1}{2}e^{-2(x-2)}$
2	$\frac{1}{4}e^{-2(x-2)}$
0	$-\frac{1}{18}e^{-2(x-2)}$

9. Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ and assume that X and Y are independent.

(a) Show that $X + Y \sim \text{Poisson}(\lambda + \mu)$

$$\begin{aligned}
 \text{Since } X \perp Y, \quad M_{X+Y}(t) &= M_X(t) \cdot M_Y(t) \\
 &= e^{\lambda(e^t-1)} e^{\mu(e^t-1)} \\
 &= e^{\lambda(e^t-1) + \mu(e^t-1)} \\
 &= e^{(\lambda+\mu)(e^t-1)} \\
 &\star \text{Poisson MGF w/ parameter } \lambda+\mu \\
 \text{So, } X+Y &\sim \text{Pois}(\lambda+\mu).
 \end{aligned}$$

(b) Find the distribution of $X \mid X + Y = n$.

Let $U = X$ and $V = X + Y$. Then $X = U$ and $Y = V - U$.

$$f(X \mid X+Y=n) = \frac{f(u,v)}{f(v)} = \frac{e^{-(\lambda+\mu)} \lambda^u \mu^{v-u} \cdot \frac{1}{u!(v-u)!}}{e^{-(\lambda+\mu)} (\lambda+\mu)^v \left(\frac{1}{v!}\right)}$$

$\star$ Setting $V = n$ and cancelling/combining terms...

$$\begin{aligned}
 &= \frac{n!}{u!(n-u)!} \lambda^u \mu^{n-u} (\lambda+\mu)^{-n} \cdot \left[(\lambda+\mu)^u (\lambda+\mu)^{-u} \right] \\
 &= \binom{n}{u} \left(\frac{\lambda}{\lambda+\mu} \right)^u \left(\frac{\mu}{\lambda+\mu} \right)^{n-u} = \binom{n}{x} \left(\frac{\lambda}{\lambda+\mu} \right)^x \left(\frac{\mu}{\lambda+\mu} \right)^{n-x}
 \end{aligned}$$

$$\boxed{\text{So, } X \mid X+Y=n \sim \text{Binom}_7\left(n, \frac{\lambda}{\lambda+\mu}\right)}$$

10. Let X and Y be iid $Exp(\lambda)$. Find the pdf of $U = \log(X/Y)$.

$$f(x, \tau) = \lambda e^{-\lambda x} \lambda e^{-\lambda \tau} = \lambda^2 e^{-\lambda(x+\tau)}$$

$$U = \log\left(\frac{X}{Y}\right) \quad V = Y$$

$$Y = V \quad ; \quad U = \log\left(\frac{X}{V}\right) \quad ; \quad e^U = \frac{X}{V} \quad ; \quad X = Ve^U$$

$$J = \begin{vmatrix} Ve^U & e^U \\ 0 & 1 \end{vmatrix} = Ve^U$$

$$f(u, v) = \lambda^2 e^{-\lambda(Ve^U + V)} \quad Ve^U = \lambda^2 Ve^U e^{-\lambda V(e^U + 1)}$$

$$0 < v < \infty$$

$$0 < Ve^U < \infty \quad ; \quad 0 < e^U < \infty \quad ; \quad \log(0) < U < \log(\infty) \quad ; \quad -\infty < U < \infty$$

$$f(u) = \int_0^\infty \lambda^2 Ve^U e^{-\lambda V(e^U + 1)} dv = \lambda^2 e^U \int_0^\infty Ve^{-\lambda V(e^U + 1)} dv$$

$$\begin{aligned} u=U \quad dv &= e^{-\lambda V(e^U + 1)} \\ du=1 \quad v &= \frac{1}{-\lambda(e^U + 1)} e^{-\lambda V(e^U + 1)} \end{aligned} \quad \lambda^2 e^U \left[\frac{V}{-\lambda(e^U + 1)} e^{-\lambda V(e^U + 1)} + \int \frac{1}{\lambda(e^U + 1)} e^{-\lambda V(e^U + 1)} dv \right]_0^\infty$$

$$\lambda^2 e^U \left[\frac{V}{-\lambda(e^U + 1)} e^{-\lambda V(e^U + 1)} - \left(\frac{1}{\lambda(e^U + 1)} \right)^2 e^{-\lambda V(e^U + 1)} \right]_0^\infty = 0 - 0 + 0 + \lambda^2 e^U \left[\frac{1}{\lambda(e^U + 1)} \right]^2$$

$$= \frac{e^U}{(e^U + 1)^2}, \quad -\infty < u < \infty$$

~~To let X and Y be iid $\text{Exp}(\lambda)$. Find the pdf of $U = \log(X/Y)$.~~

OR

$$\frac{1}{e^u \left(1 + \frac{1}{e^u}\right)^2}, \quad -\infty < u < \infty$$

OR

$$\frac{1}{e^u + 2 + e^{-u}}, \quad -\infty < u < \infty$$

Depending on parameterization and level
of simplification

11. If P is a probability function and A and B are any sets in some σ -algebra $\mathcal{B}$, then prove the following:

(a) $P(B \cap A^c) = P(B) - P(A \cap B)$

Note that $P(B) = P(A \cap B) + P(A^c \cap B)$ by
Law of Total Prob.

$$\begin{aligned}\Rightarrow P(B \cap A^c) &= P(B) - P(A \cap B) \\ &= P(A \cap B) + P(A^c \cap B) - P(A \cap B) \\ &= P(A^c \cap B) \\ &= P(B \cap A^c)\end{aligned}$$

(b) If A and B are independent, then $P(A \cup B) = 1 - P(A^c)P(B^c)$

Note: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Note: If $A \perp B$, $P(A \cap B) = P(A)P(B)$

$$\begin{aligned}\Rightarrow P(A \cup B) &= P(A) + P(B) - P(A)P(B) \\ &= P(A)(1 - P(B)) + P(B) \\ &= P(A)P(B^c) + P(B) \\ &= P(A)P(B^c) + 1 - P(B^c) \\ &= -P(B^c)(1 - P(A)) + 1 \\ &= 1 - P(B^c)P(A^c). \quad \square\end{aligned}$$

11. If P is a probability function and A and B are any sets in some σ -algebra $\mathcal{B}$, then prove the following:

b)

(a) $P(B \cap A^c) = P(B) - P(A \cap B)$

OR

$$\begin{aligned} P(A \cup B) &= 1 - (P(A \cup B)^c) \text{ by complement rule} \\ &= 1 - P(A^c \cap B^c) \text{ by De Morgan's Law} \end{aligned}$$

Note: Because $A \perp B$, we know $A^c \perp B^c$ from homework problem (or you could prove it here).

(b) If A and B are independent, then $P(A \cup B) = 1 - P(A^c)P(B^c)$

$$\Rightarrow P(A^c \cap B^c) = P(A^c)P(B^c)$$

$$\begin{aligned} \text{So, } P(A \cup B) &= 1 - P(A^c \cap B^c) \\ &= 1 - P(A^c)P(B^c). \end{aligned}$$